

Question-1 (the student can review lecture 1&2)

Graph the following functions and find its domain and range.

$$1) \quad f(x) = \begin{cases} \csc x & x < -\pi \\ \tan x & -\pi \leq x \leq \pi \\ \sec x & x > \pi \end{cases}$$

$$2) \quad f(x) = \begin{cases} x^2 & x < 0, \\ 2 - \sqrt{x} & 0 \leq x \end{cases}$$

$$3) \quad f(x) = \begin{cases} \sec x & x < -\frac{\pi}{2} \\ \tan x & -\frac{\pi}{2} < x < \frac{\pi}{2} \\ \csc x & x \geq \frac{\pi}{2} \end{cases}$$

$$4) \quad f(x) = \begin{cases} 2 - (x+3)^3 & -5 \leq x < -3 \\ \sqrt{x+3} + 2 & -3 \leq x \leq 6 \\ \frac{1}{x-6} + 8 & x > 6 \end{cases},$$

$$5) \quad f(x) = \begin{cases} (x+3)^3 + 2 & x \leq -3 \\ 2 - \sqrt{x+3} & -3 < x < 6, \\ (x-6)^2 + 5 & x \geq 6 \end{cases}$$

$$6) \quad f(x) = \begin{cases} -x^3 & x < 0 \\ 2 - \sqrt{x} & 0 \leq x \leq 3 \\ \frac{1}{x-3} & x > 3 \end{cases}$$

Question-2 (the student can review lecture 2)

Show that:

- 1) $\tanh^{-1} x = \frac{1}{2} \ln \left| \frac{1+x}{1-x} \right|$
- 2) $\cosh x + \sinh x = e^x$
- 3) $\coth^{-1} x = \frac{1}{2} \ln \left| \frac{1+x}{x-1} \right|$

4) $\sinh^{-1} x = \ln \left(x + \sqrt{x^2 + 1} \right)$

5) $\cosh^{-1} x = \ln \left(x + \sqrt{x^2 - 1} \right)$

Question-3 (the student can review lecture 4)

- 1- Find **Maclaurin series** for the following function. $f(x) = 9^x$, hence find the series for $f(x) = x^7 9^x$.
- 2- Find **Maclaurin series** for the following function. $f(x) = \cosh x$, hence find the **linear** approximation and **Quadratic** approximation.
- 3- Find **Maclaurin series** for the following function. $f(x) = 5^x$, hence find the series for $f(x) = x^7 5^x$.
- 4- Find **Taylor series** for the following function. $f(x) = \ln \sqrt{x^3}$, about $x = 1$ hence find **linear** and **quadratic** approximations.

Question-4 (the student can review lecture 3&4)

Find y' for the following functions.

(1) $y = (\tan x)^{\sinh^{-1} \sqrt{x}}$

(2) $y = \pi^{(\cos^{-1}(\tan x))} + \ln \sqrt[3]{\sec^4(e^{2x})} + \sin^{-1}(e^{\sinh x})$

(3) $3xy^2 + x \cosh y^2 = 5 \cot^{-1} y$

(4) $\tan y - y^2 \cosh x + x^2 = \sinh x$

(5) $y = (x)^{\sin x}$

(6) $y = 9^{(\sin^{-1} x)} + \ln \left(\sec^2(e^{2x+3}) \right) + \sqrt{\cot(x^2 + 5)}$

(7) $y = (\tanh x)^{\sinh^{-1} \sqrt{x}}$

(8) $y = 45^{(\cos^{-1} x)} + \ln \sqrt[3]{\sec^4(e^{2x})} + \sin^{-1}(e^{\tan x})$

(9) $3xy^2 + x \cos y^2 = 5 \cot^{-1} y$

(10) $y = (\tanh x)^{\sin^{-1} \sqrt{x}}$

(11) $y = 45^{(\cosh^{-1} x)} + \ln \sqrt[3]{\sec^4(e^{2x})} + \cos^{-1}(e^{\tan x})$

(12) $y = (\cos x)^{\sinh^{-1} x}$

$$(13) \quad y = 9^{(\tan^{-1} x)} + \ln \left(\sec^3(e^{2x+3}) \right) + \sqrt{\tanh(x^2 + 5)}$$

$$(14) \quad 3xy^2 + \cos y^2 = 2x^3 + 5$$

Question-5 (the student can review lecture 6)

Evaluate the following limits:

- (1) $\lim_{x \rightarrow 0} \frac{\sin x^2}{\ln(\cos x)}$
- (2) $\lim_{x \rightarrow 0} \frac{e^x + e^{-x} + 2}{e^x - 1}$
- (3) $\lim_{x \rightarrow 0} (x + \cos x)^{\frac{k}{x}}$
- (4) $\lim_{x \rightarrow 0} \left(\cot x - \frac{1}{\sin x} \right)$
- (5) $\lim_{x \rightarrow 0} \frac{2e^{-88x} + e^{-10x} - 3}{e^{5x} + \cos x}$
- (6) $\lim_{x \rightarrow 0} \left(\frac{1 - \cos x}{3x^2} \right)$
- (7) $\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right)$
- (8) $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$

Question-6 (the student can review lecture 5)

- 1- Find the critical points of $f(x) = 2x^3 + 3x^2 - 36x$, intervals at which $f(x)$ is increasing, decreasing also find the local max & local min points. Find the inflection points and the intervals where the function $f(x)$ is concaved up and where it is concaved down.
- 2- Find the inflection points and the intervals where the function $y = x^4 - 4x^3 + 10$ is concaved up and where it is concaved down.

3- Find the critical points of $f(x) = 2x^3 - x^4$, intervals at which $f(x)$ is increasing, decreasing also find the local max & local min points.

Question-7 (the student can review lecture 7)

Resolve into partial fractions.

1) $f(x) = \frac{5x - 20}{x^3 + 5x}$

2) $f(x) = \frac{x + 3}{(x + 1)(x^2 - 5x + 6)}$

3) $f(x) = \frac{2x^2 + 6x - 4}{x(x + 2)^2}$

Question-8 (the student can review lecture 8&9)

A- If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$

1- Prove that $A^2 - 4A - 5I = 0$,

2- Find A^{-1} .

B- IF $A = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 2 & 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 2 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 8 & 7 \\ 0 & 7 \\ 1 & 0 \end{bmatrix}$ Find the matrix $A^{-1} + BC^T$.

C- Solve the following system of equations, using a suitable method.

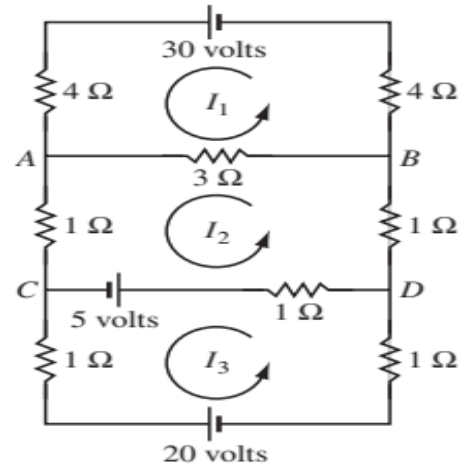
$$x_1 + x_2 + x_3 = 3$$

$$2x_1 - x_2 + 4x_3 = 1$$

$$-x_1 + x_2 - x_3 = 0$$

D- Find the electric currents shown in the following circuit by solving the system using two different methods.

$$\begin{aligned} 11 I_1 - 3 I_2 &= 30 \\ -3 I_1 + 6 I_2 - I_3 &= 5 \\ -I_2 + 3 I_3 &= -25 \end{aligned}$$



With best wishes Assoc. Prof. Dr. Mohamed Ali